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CMPE 277
Hackathon Innovation: Scalable Voice-Driven Mobile Apps for Regulatory Compliance and ESG Impact

After graduation, you joined a high-profile consulting firm that partners with multiple industries, with strong focus areas in ESG (Environmental, Social, and Governance), econometrics, and regulatory compliance [1,2]. To accelerate innovation and respond to rising demand for scalable, enterprise-grade mobile solutions, the firm launched an internal **hackathon-style initiative**—aimed at building a **next-generation, touchless mobile app experience powered by voice navigation**.

You were selected as a core contributor and assigned two high-impact challenge tracks:

1. An ESG-focused solution leveraging real-world datasets (including food security insights), and
2. A mission-critical **California DMV Knowledge Test mobile application** designed for compliance-driven industries.

Operating in a fast-paced, innovation-driven hackathon environment, your objective was to rapidly prototype and deliver a scalable mobile solution that meets strict state compliance requirements. The company's tours, travel, and on-demand charter services depend on ensuring that employees stay up to date with continuously evolving DMV rules. This created an urgent need for a **smart, accessible, and continuously updated learning platform**.

With access to key data sources—including ESG datasets and the California DMV Knowledge Test—you were tasked with building an **end-to-end mobile application** that not only ensures compliance but also reimagines user experience through voice-first interaction and intelligent automation.

Your solution was expected to deliver the following capabilities: Attached PDF and uploaded to Class Canvas.

Hackathon: March 28, 2026 (11:00 am PST)– March 29, 2026 (12:00 noon)
Session: CMPE 277 - Smartphone Application Development, Spring 2026
Academic Career : Graduate
Specialization: MS Software Engineering, Specialization in Cloud and Mobile Computing
Instructor: Professor Chandrasekar Vuppalapati, chandrasekar.vuppalapati@sjsu.edu

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References

1. C. Vuppalapati, *Machine Learning and Artificial Intelligence for Agricultural Economics: Prognostic Data Analytics to Serve Small Scale Farmers Worldwide*. Cham, Switzerland: Springer, 2021
2. C. Vuppalapati, *Building Next-Gen ESG Platforms with IoT and AI for Sustainable Development Goals*, 1st ed. Boca Raton, FL, USA: CRC Press, 2026.

Mockups

Uploaded to Class Canvas

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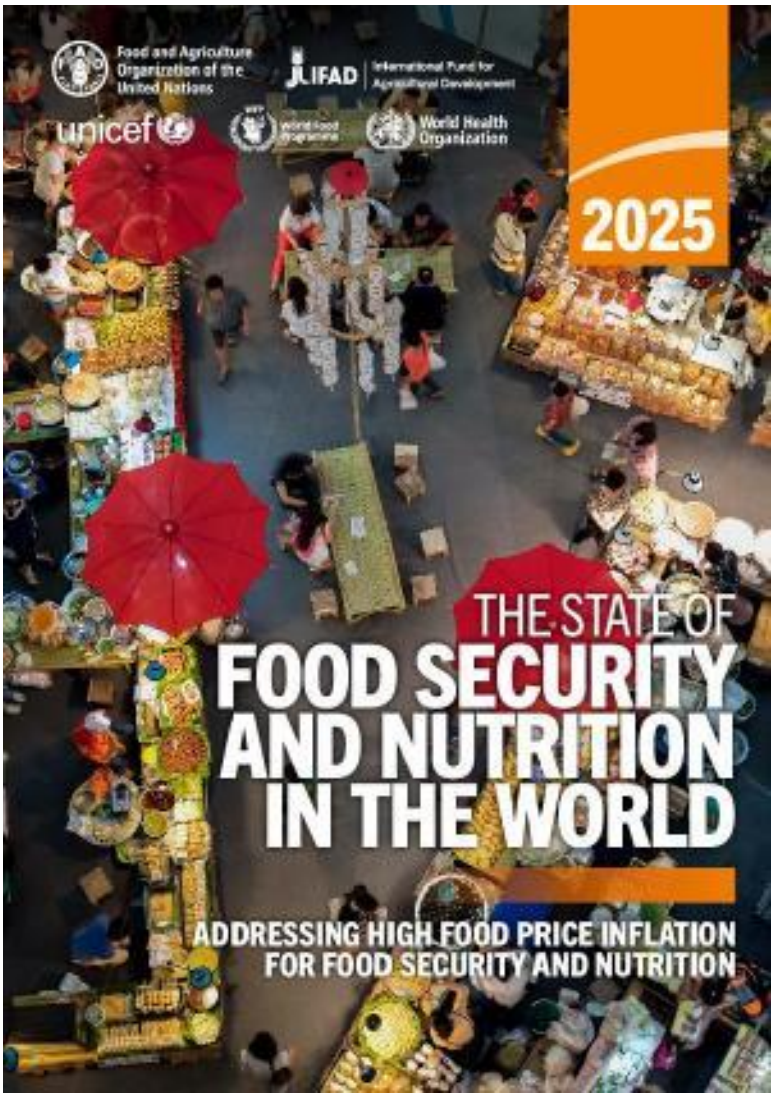
Data Sources

The State of Food Security and Nutrition in the World 2025

https://openknowledge.fao.org/items/4b1f7d26-267d-4a81-aed4-4f9de4d93f85



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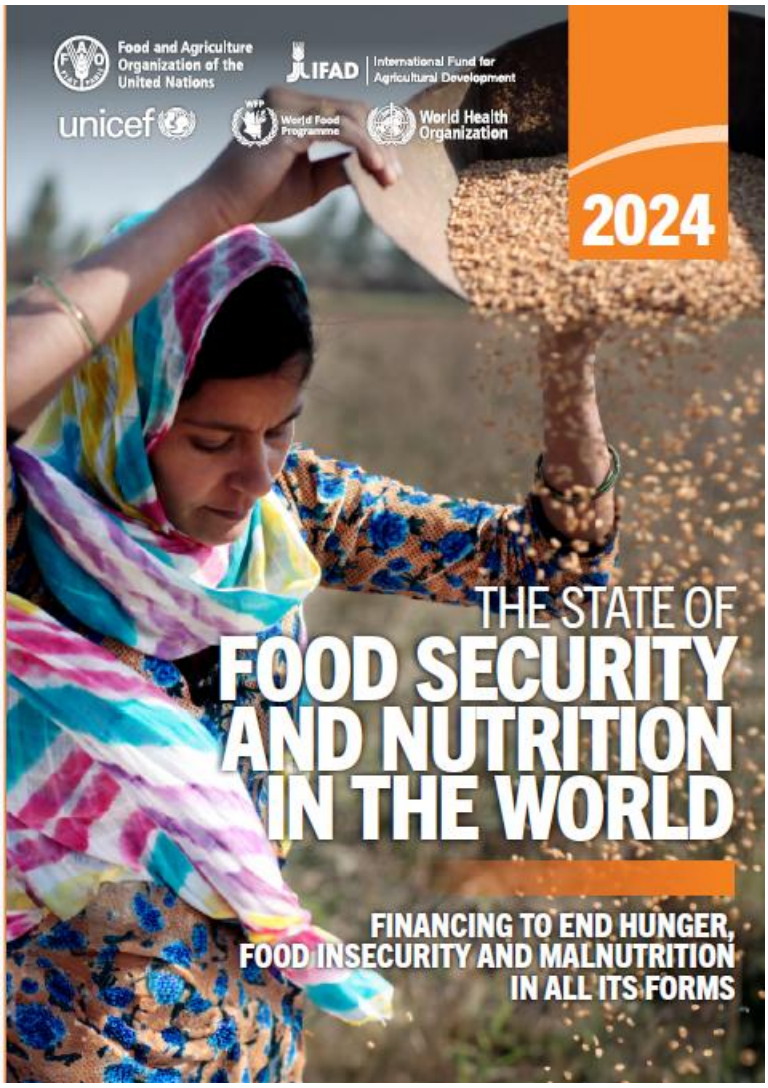
Data Sources: The State of Food Security and Nutrition in the World 2024

https://www.who.int/publications/m/item/the-state-of-food-security-and-nutrition-in-the-world-2024



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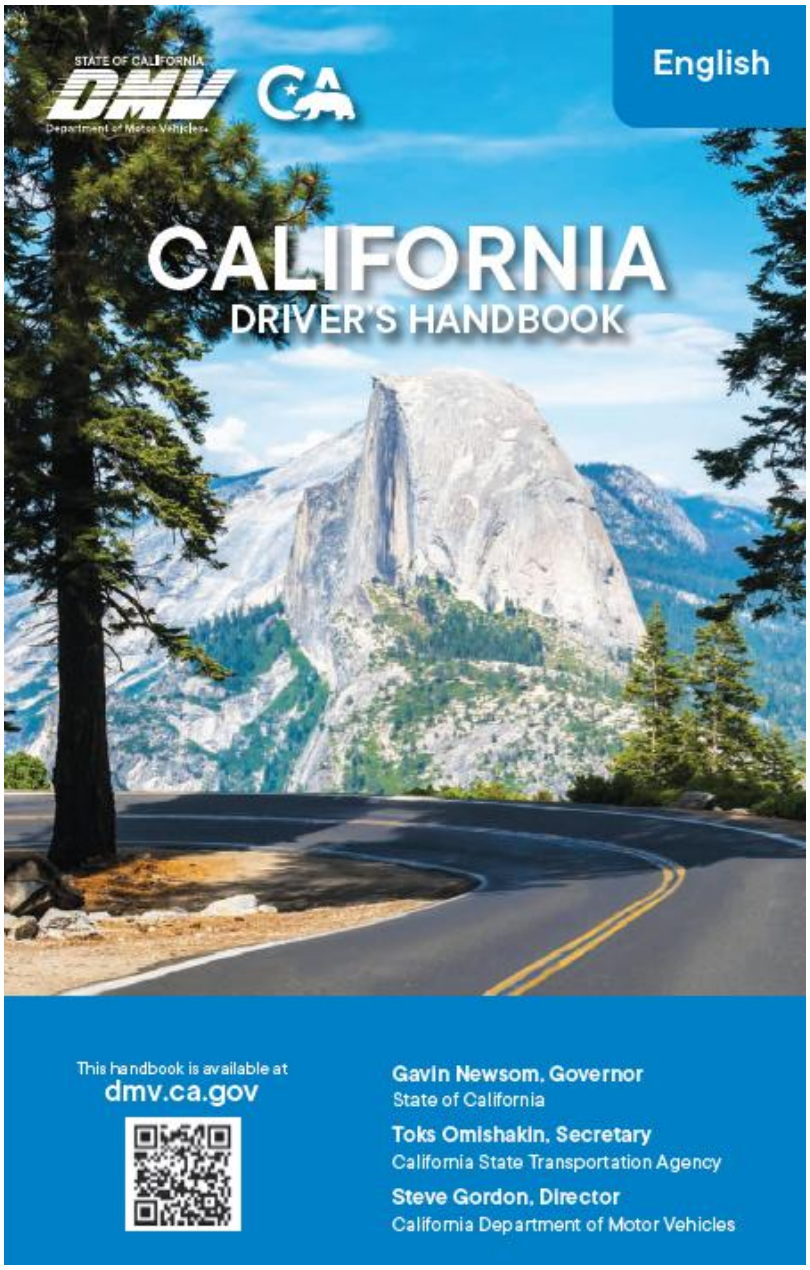
California Driver's Handbook

https://www.dmv.ca.gov/portal/file/california-driver-handbook-pdf/



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